ActInf GuestStream #015.3 ~ Bobby Azarian: "The Teleological Stance: Th

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Hello and welcome everyone. This is ACTIMPF Guest Stream 15.3. Lucky number three with Bobby. We're here with Bobby Azarian as well as some guests. We're going to have a discussion, talk about a range of different topics. It's June 1st, 2023. First, Adam and Michael can introduce themselves. Then Bobby will pass it off to you for sharing your screen and facilitating this however you see fit. So Adam, go for it. Hello, I'm Adam. I am a research fellow at the Johns Hopkins Center for Psychedelic and Consciousness Research. Long standing interest in the free energy principle, consciousness, free will, psychedelics, artificial intelligence. And along the way, I met and befriended Bobby and I guess proud to be part of the movement. So thanks for coming, Adam. Yeah, Adam, I wish we would have time to get into this today, but Adam has presented a pretty clear mechanistic model of free will, cybernetic free will. And there's just this huge problem where, and I'll keep this short, we're going to introduce Michael, but basically the main kind of straw man against free will is that there's this argument that you have this non-physical mind doing like stuff outside of causality. And that's why everyone's rejecting it as not a scientific idea. But of course, something can't work if it's not, doesn't have mechanism. If it's not real, it's not real. So it's kind of a ridiculous debate to have. If you want to talk about what people are talking about when they discuss free will. In the past, we didn't have mechanisms to understand what that was, but now we understand things like top-down causation. We understand control. And when you have that language, you can actually talk about what free will is in a mechanistic way. And that just changes the debate completely. That's not enough to convince people that there's something called free will because there's a lot of issues that we get into. But Adam's one of the people that have been taking this nebulous concept and have been trying to show that there's actually a very specific set of mechanisms and it's nothing outside of science, but it is science that is outside of our current paradigm. It's just dynamics that we don't normally, that weren't included in the previous model. So it looks like magic, but it's actually, there's stuff going on there. So yeah, we're going to have a longer discussion with Adam. In the future, either on my channel, or maybe we'll come back and make it all about free will because it's such a big topic. But yeah, thanks for coming.
And invite Michael as well. So, yeah, you want to,
Am I supposed to take the wheel now? Yeah. Obviously I'm a fan of Bobby's. I very much like, well, I think it's important. I think the work that Bobby is trying to articulate and Adam, I'm a fan. So this is the first time we've met, but I've watched some of your stuff as well. And Daniel, obviously we've talked. And yeah, let's just yield to Bobby and let's see where we go here. Yeah. Michael has a background in kind of system science and those kinds of things. That's how we kind of resonate is kind of like the system science approach, but thinking about how system science can be applied to like psychology and organizational issues. So how to make this stuff that's very

theoretical, practical. And so we're kind of working on that on the side. And then Brendan
Grand MC will be coming and joining the conversation since he's not here right now. I'll just give a
little introduction for him. He's the author of a book called Emergentism, and he has a Facebook
group called Metamodern Spirituality. And basically he's in that book, he's turned this emergence paradigm,
which is basically kind of the opposite of the reductionist paradigm, into something like a sort
of secular religion. And so he's talking about a lot of these same things. So he'll join us shortly,
probably. I think he's watching the stream and is trying to get into the Zoom link.
So yeah, we can start. It'd be great for this to be sort of a discussion. So if anything
I say triggers any sort of thoughts or criticisms, please speak up because I don't have an organized
set of slides. And there's a lot to talk about. And the problem really when I was trying to make
these slides was like everything I say kind of gets into something else. And so I had like a
hundred slides and I was just trying to organize them in the last minute. So I think, you know,
I have some slides that were made in the past for other presentations that are just going to be
kind of like a roadmap. But we can veer off of that. And I'm going to be saying a lot of stuff
that's not in the slides. Yeah, so I guess we can start. So I'll share my screen because I just shared
my screen for my computer and I couldn't see you. But I think the trick to that is to share my screen
through Zoom and then share do like slideshow and then I should be able to see your faces on the side.
Yep. Okay, so you can see my screen now. Yep. And then if I do slideshow, it doesn't cover you up.
Perfect. Yeah. Perfect.
Okay. Um, so
Bobby, just before you get started, I believe Brendan's joined us as well.
Hey, Brendan.
We warmed up the crowd.
His audio is probably not connected yet. But um, yeah, so we can just begin.
Uh, hey, Brendan, I introduced you and uh,
Yeah, I was just saying YouTube version and I got the link. So great. It's a pleasure to be here.
Thanks for the introduction. Appreciate it.
Yeah, no problem. Please feel free to like chime in at any time because I really need help here.
I'm not like prepared to do this by myself. So, um, and maybe you can play the role of the critic too,
because I should have invited some like anti free will people because, um, or just some skeptics of
these ideas because what I realized is like, I'm saying these things, a lot of them were the things
in my book, the romance reality. And I make that argument as clear as I can, but then people have
criticisms that for whatever reason, you know, how I worded it, don't address those. And then they're
not convinced. And then I do nothing to convince anyone who doesn't already believe this. So it's
actually a lot more, um, fruitful if there's a sort of this dialectical debate, like this dynamic,
because then you hear both sides and then I can actually address the things that people think are the
issues that might, um, you know, cast out on some of the things that I'm arguing. So, um, the teleological
stance. So this is the name of the self-help system for a lack of a better, uh, term. I know people

hear self-help and they think, uh, you know, that's something like cheesy and new age or, you know, BS. And there are a lot of like, you know, snake oil salesmen in this place. And I think that's why it's so important that we try if possible to create something like a self-help system and, you know, something almost like a religion, like something that has practices and a sort of instruction manual for living, um, based on scientific principles. Now it doesn't have to be based on, you know, science won't explain everything, um, or address everything. Um, but this provides a foundation where we can actually start to build a logic where we can go. Some of these things are right. And some of these things aren't, and we can start to see things that have been said in folk psychology or new age practices that people just considered to be non-scientific or pseudoscience. We can see how that maps on to some of these newer mechanisms that we're understanding because now we're starting to look at biology and neuroscience and psychology from the perspective of information theory and thermodynamics. And these are like revealing new mechanisms that I believe can create a bridge between reductionist science and the higher sciences, but also a lot of these statements you'll hear, you know, things that you hear in like spiritual circles, or people are doing practices or, uh, and, you know, just different little things that they find to be very effective. In most cases, there's a scientific explanation for why those things are effective. And most science-minded people are ignoring all that stuff. So they're like throwing out all of these things that emerged because they were adaptations like practices, um, you know, community building, things like that are just completely being ignored. Um, and that's really bad. It's just throwing the baby out with the bathwater. And as long as we have that mentality, um, you know, scientists, they may say, you know, I don't want to promote religion, organized religion, let's say, or, you know, like kind of new age spirituality. I don't want to promote that, but they're actually helping keep those things in existence because the, a lot of regular people are hearing this scientific worldview. That's the mainstream worldview. And it's very nihilistic. And some of its beliefs, and that'd be okay if that's how reality was, but we're going to see that a lot of these are based on mistaken assumptions and people are rejecting that worldview because it's doesn't offer anything of value to them as a normal person. And they will opt for other less evidence based, uh, worldviews or philosophies because they're not getting that from that. So science is actually helping like pseudoscience and things like that. Um, if they want to keep science isolated from all of these practical things and sort of addressing bigger existential and even spiritual questions.
And so if we can do that, if we can build that bridge, it would be good for unifying everyone, but it would also cut down on a lot of the, the pseudoscience and stuff that people are trying to, um, kind of prevent from, you know, not going there. But actually I think that's, um, part of the problem, if that makes any sense. Um, so this is stuff from my Substack Road to Omega. Go there and check it out. There's, uh, posts on the teleological stance that break this down in, um, more detail than I'll be able to give here today. Um, I have this word in this slide, poetic metanaturalism. If you're familiar with my book, um, you might've seen that it's not really important, um, to really talk about why I call it that, but let's just say there are these two distinctive worldviews. So there's the old mainstream scientific worldview, which is the reductionist worldview, which I think is basically a nihilistic worldview. Um, and, uh,

there are a lot of assumptions built into this worldview based on just, we had limited knowledge at the time of how the world works. So there were things like, for example, when we thought the world was Newtonian, you just couldn't get away from determinism. Uh, this kind of strict Laplacian determinism Laplace basically took Newton's, um, kind of framework and tried to apply it to everything. It's really interesting that Newton himself didn't do that. He didn't think that his laws applied to life and Newton was actually very religious and thought there were all these crazy hidden messages in the Bible, which is neat and kind of like covered up because people, you know, don't want people to know that, that Newton was so religious. But, um, so Laplace kind of, you know, created this deterministic philosophy. And when we thought that was the truth, then there was no room for human agency. There was just one future trajectory and we couldn't do anything to change that. And we couldn't, we weren't really doing anything to begin with. It was all, everything we do is sort of illusory and we can talk about determinism in greater detail. Um, but that's one nihilistic aspect is that we don't have any agency and we don't have free will free will. It's a bit more of a loaded term.

There's just assumptions with that word, like with the word free that, uh, make it a little bit more of a contentious subject, but, um, yeah, without, with determinism in the strict sense, no one could have any agency. So it kind of, if you accept that scientific, um, proposition, then you're like basically accepting that you have no control over the future or your life. And that can be very depressing and demotivating. And there are studies that we can talk about that have shown that, um, that belief actually causes people to, for example, um, behave less morally. They will be more likely to cheat, uh, because they don't have culpability because no one's responsible for anything. That's them basically choosing to not have this agency that I'll explain exactly what I mean by that. But basically, I mean, the fact that you can tell them that, and then they lose it and then they start behaving more immorally, it's obvious that there's something causal there that's changing their behavior. So that's kind of a clue that that that picture, something doesn't make sense there. Um, and it also causes people to be like really depressed and have kind of existential meltdowns. Um, I personally had, you know, when I thought determinism was the only explanation that made sense, I sort of had, you know, times where I would have these experiences where I was very disturbed by the fact that I had no control over anything and that this was sort of a tape that was determined from the initial event and that I was just playing that out. It can be very disturbing if you take this stuff seriously. And what I'm arguing is that a lot of the scientists preaching this stuff, they're not taking it seriously because they can't, it would be, your life would just be over if you really believe that you had no agency because you would stop doing stuff. Um, I tried to take it seriously and, uh, it took me out for a good two years. Um, how do you even take it seriously? Like, what does that do? What do you do then? Um, so, so the, it happened, um, it was a combination of reading the, this book, um, the meme machine, um, by Susan Blackmore and, um, Darwin's dangerous idea. There's this one sentence that just like took me out. It was like, um, uh, everything up to this moment, including your genes, environment, uh, and all your experiences, including the first half of the sentence will determine how you, uh, interpret

the second half. It was sort of like just this web of causation being described on a different level of analysis, sort of like bringing that to the foreground. And I was like, and not seeing me in that foreground. And then I was like, oh, wait, there isn't like a neuron in my brain that knows or cares that I exist as whole or, or, and then it was wait, not even a wait. So sodium ions are gonna sodium ion. They're just gonna follow those gradients like, oh God. And I, so, um, and it just sort of had me. And, uh, and so for, I just was, I mean, so I was actually, so for a while, I was, um, uh, uh, is it kind of paralyze you or did you just go, I'm going to do what I was going to do anyway, but I'm going to do it. I'm going to have this weird sort of, you know, experience at the same time because it's all meaningless. I'm just going to go through life. I'm probably going to look really sad, but I'm going to do everything that I plan to do otherwise, or does it just shut you down where you're like, I'm not going to get out of bed. There was kind of like a double consciousness about it too. Cause it was like, I'd be moving between like, um, like, having that as a, as a frame and, but then like to take action, you're like, you can't, you can't, you just gotta throw it out the window. And so like, yeah. So, and some, on occasion, there would be kind of like, uh, um, maybe this is a, I don't think it took me in an unethical direction, but, um, kind of like a, like a fracket mindset, like, uh, well, you know, nothing matters anyway, so woo, but yeah. And it really stick for me. Can, can I offer a devil's advocate take here? And I don't know if we're, I don't know if we're derailing or no, this is perfect.
Think of it. So, uh, this is great. But, um, so, and it is very much a devil's advocate position. Cause I think Bobby, you and I are very much aligned on trying to emphasize the importance of, uh, free will, the, the causal power of information, potentially, et cetera. I'm sure you might get into some of that. Uh, but I will say this in its favor or in, in addition to what's being said.
Um, so I also had an experience like that of suffering from the, uh, you know, seeming implications of the deterministic universe. But I do think that there's a way of also allowing all of that to be the case and responding differently. And there are a number of different ways that that can look. Um, I mean, there's the Nietzschean approach to this, which is actually kind of taking this deterministic cycle and kind of the eternal recurrence of the same as sort of a wild yes saying to existence. Um, and I'll just, I'll just throw that out there and bracket it. We could come back to that one. Uh, the other thing by that, that's sorry. I don't know.
So Nietzsche was, he would philosophize while walking, or he would think his philosophy while going on these long walks and everything. Right. And one day he was walking in the Swiss Alps and he sees this gigantic rock. Um, and, uh, he comes to this sudden realization. Um, and I guess I won't get into the whole story cause it would probably be a little bit too much, but basically the, the, the realization that springs there, which he, uh, which he has when at this sort of like monolithic scene is somehow he's sort of transported and appreciates that, um, what, what he would need to do was affirm his existence, uh, under the assumption that everything that has happened and will happen, uh, will happen over and over and over again, uh, for eternity. Uh, all the mistakes, all the ecstasies, et cetera. It was all there, uh, almost, you could kind of think of it kind of in

the, uh, you know, there's the big bang, big crunch model, and then it repeats. And so
let's say something like that is happening where it just repeats over.
Yeah. He called it eternal recurrence. Right. I mean, yeah. Yeah. Yeah. Yeah.
A eternal recurrence. And he basically said that, um, we need to be able to positively affirm that
in an exuberant sense. Uh, and, and actually, and here's, I think the point,
what it means is that if you take the idea seriously, um, that this is your life and you
are therefore radically responsible in some ways for what you experience, um, uh, and I'll come back
to this in the sense of what responsibility could mean in this context as well. Um, then,
then it actually imbues everything with an incredible significance because this is what you'll be
experiencing over and over again. Uh, the second thing I wanted to say, which I think might kind of
preempt some of your kind of like, well, how does that work? Right. How can you be responsible?
Uh, is, um, I do think that there's a phenomenal logical way that you can, uh, come to terms with
determinism, uh, that basically concludes something like, okay, even if mechanistically,
this can be describable in terms of deterministic cause and effect all the way back. Um,
my experience of it is rather different. I have the experience of free will.
I feel as though, as if I can choose as, as I want to. Uh, and so there's no necessary reason
why some descriptive, uh, you know, framework needs to, uh, needs to impinge upon my own
phenomenological sense of reality. Uh, so for a long time, I made peace with this question that way
by basically saying, yeah, it's all determined, but I feel as though I have free will. Um, and that's
really, I think the, the, the core there because, um, let me, yeah, let me say something about that.
And I don't want it to be seem dismissive. Um, but I mean, this is not your idea anyway,
so I don't really have to apologize to you. It's Nietzsche. And it's one of the possible things that
I think you're considering, but I mean, I, I, you know, we've talked about this. I think you kind of,
we have similar ideas about agency. What I want to say about that is basically that is similar to
compatibilism and what's going on is that these people are, there's language games. First of all,
like if you want to talk about that, if you believe that causal story in the strict sense,
um, and it's more complicated that because I'm saying that the, the, there is causality,
there's always cause and effect to everything. And that doesn't not permit agency. But what I'm
going to say is the, the, the kind of magic ingredient to understanding how this can be,
uh, you know, uh, mechanistic idea, um, or, or mess in a mechanistic process, how free will can be
explained as part of the causal chain. Uh, we need to talk about top-down causation and that's really the
game changer. And, but without top-down causation, I don't think there is any responsibility. And what
I think that Nietzsche and others are doing is they're, first of all, they're compartmentalizing.
They're like, they're, they're, they're not fully taking the determinism they're talking about
seriously. And if you do that, like you, you have to compartmentalize if you believe that,
because like Adam said, like, you have to throw that out when you actually want to take action.
So first of all, everybody who believes in determinism is kind of living a split personality
life. And that, that leads to all kinds of cognitive dissonance, but it's worse than that.

All of all, I mean, you still are.
I mean, in a way it depends. So like, um, I'm agnostic about determinism in terms of like,
it was, that was part of, but it wasn't agnosticism about determinism, but going more deflationary on causation or more instrumentalists on causation. And I do think that, but I'm open to there being like a best causal handling, or like, there could be like a new, like, um, for me, where like causation kind of like enters into physics and like full majesty. And like,
Adam Werner, that's what this is. That's what I want to describe.
Adam Werner, But the move for me was, um, compatibilism via instrumental causation that kind of stays within as, as like, Carol's like, it stays in its lane.
Adam Werner, Yes. And I, so I will put out a thing out there where you could have determinism and have true agency to at these higher levels, you would still need top down causation, but you might not need any indeterminism at the most fundamental level, uh, like quantum indeterminism.
Adam Werner, Um, and you could still have, uh, potentially agency and that's sort of Adam's perspective, or like he says, he's agnostic on it. But, um, in the future, I hope to convince Adam that there's a better solution. And so, so what I was saying to Brendan is that these are language games when he's like, uh, it's truly this way, you know, it's this way, but that should make you feel like you're responsible because this is, you know, the life you have and the one you're going to repeat or whatever, you know, I'm it's good reasoning. Brian Greene says similar stuff.
And that's why I say this Brian Greene's like, the decision is real, but it's not really you making it, but it's a real, like a lot of strange, weird language games that if you actually make language, the focus, which I wanted to do, but it's way too abstract. Like I didn't have time to nail that down for today, but that's what I want to do in the future. And you can show that people who are talking about this stuff, especially people who are, you know, trying to find some sort of silver lining to determinism, they're, they're using words inconsistently. They will use a word one way and then contradict it. They might even use the same word in two different ways, um, in the same sentence, but basically you can't have, well, I'm going to let you respond, Brendan, but I don't think there can be in his, that, that model, any true responsibility. And then this solution of like, but I feel like I have it to me, it doesn't give you anything additional. Like you're just an epic phenomenon.
And to me that can be almost worse. Like he's given a nice spin on the positive way you can see it, but there's other people like illusionists that say, yeah, we have this phenomenal experience.
It feels like I have free will, but there's absolutely no responsibility. Like Sam Harris was like, I don't have pride. Uh, a friend was just talking to me about this cause he's a Sam Harris fan, but he's turned off by this whole no free will thing. And then he hears Sam Harris contradict himself all the time because Sam Harris will say, you don't have free will. And the whole thing is on free will. And then there's a little commercial for his podcast where he goes, you have the freedom to choose to listen to this podcast. And it's like, you're, you're using double talk like constantly. So I think it's very notable. Sam Harris does not tell that to his daughter.
Sam Harris and Annika, they do not, they say it to each other, but he says, I'm not,

I'm not going to tell my kids because it's going to make them, you know, behave differently. And like all of these bad things that I said, but when they grow up, I'll tell them that they don't have any agency when they're ready for it. But what I'm saying is all of that is absolutely ridiculous talk. It's absolutely nonsense to be like, she doesn't have a, I'm not going to like everything he's doing is like the whole arguments completely self-defeating in every way. And there's a lot of people that have tried to make nice stories out of it, but ultimately they're contradicting themselves and compatible, compatibilism in its current form is a contradiction. And people call it out for that all the time. Um, but that doesn't stop the compatibilist because it seems like the best answer. It goes, yeah, but I experienced this agency, so it has to be real on some level, but then how can we go against this determinism? It seems to go against causality in general. So it's like a lot of people like Dan didn't, their solution is like, you know, I'm not going to go against, um, the, these sorts of like, you know, I'm not going to go against, uh, determinism or causality, which I'm saying it doesn't go against causality, but they think the best solution is to try to do some magic and make both these things work together. Um, yeah, so we can talk more about that, but yeah, maybe you have a response that please. Yeah. Yeah. I was just going to throw in, um, you know, something that we had grounded this in before was the idea that, uh, you know, we choose not to act on, on the back of that. Right. So there's that balance of, you know what I'm talking about, right? Yeah. Yeah. If you, if you believe that it's going to affect your behavior too. And what I'm going to argue is that you won't exercise your agency as much as you would, uh, if you don't believe it. Um, along those lines, part of what really concerns me about the, um, anti-free will rhetoric is cause like an agency, like, so like active inference that like Carl sometimes say, like, you know, there's a certain optimism that's required, like that's just built into it for the way you pursue goals by, um, predicting them into being. And so like, if you don't, if you're not, if you stop believing in yourself as a cause, being able to predict certain goals that actually could be a real problem. I believe it is a real problem. And that actually these people living this are already not behaving optimally, not making decisions optimally. Here's the thing that you may say, Sam Harris is a really successful guy. How do you explain that? It's because he's living too, he, he, he has a split personality at the part of him that believe like, like pumped about doing all this stuff. It's disconnected from the part of him that really believes that he has no agency. Um, and you've seen that in those studies where they will like, you see people lose that belief and it changes their behavior. So my point is if, if that's the way reality is, then we have to tell the truth. And that's what Sam Harris says. Like, I'm not going to sugarcoat it for you, but what I'm saying is that idea is built on a mountain of language ambiguities that are creating this problem. Um, just a misunderstanding of, of causation, because if you think all causation is bottom up, which means that the system is not, there's no higher level control. Um, and this is kind of getting into details that I need to give some more background for, but if it's all bottom up causation, then there is no room for agency. But if you have top down causation and the whole system is this control center, um, it's a cybernetic control system that's basically, um, um, we can explain this later, but what I will argue is that the evolutionary process, um, accumulates knowledge, uh, so adaptive information in, um, biological systems. So predictive information about the environment, the knowledge, the adaptations that are being acquired through evolution correspond to, um, genes and, um, you know, neural connections that, um, uh, basically encode information about the external world. And that's what starts to give a system agency. Um, and that's why living systems behave differently than inanimate systems. And, um, um, that is because the system has gained causal power. So the system as a coherent unit, um, is starting to control the components that it's made of and the trajectory that the system follows. Um, and so for example, Philip Ball is a journalist who's been writing about this a lot, like one of the most respected science journalists. But if you drop a pigeon and a rock off the tower, the rock, you can predict with Newtonian, uh, mechanics, you can with, with, with classical physics equations, you can predict the trajectory of the rock. You can't predict the trajectory of the bird. So there's something fundamentally different about those two things. And I'm saying that behavioral distinction is what we have to call agency. Now that doesn't give us, get us to free will, but it shows us that there's some sort of causal difference between biological systems and inanimate systems. And this used to be attributed to vitalism. What I'm saying is that vitalism wasn't necessarily wrong. We just didn't know what that vital force was. And now we have a name for it and we call it information. Um, so I kind of veered from the point maybe, but, um, yeah, what, what was, uh, I can't remember what I was answering with that. Um, Brendan, do you have anything else to say? Sorry, Brendan. Yeah, go ahead. Yeah. So maybe we should finish up the Nietzsche thing and then I should get into some slides and then we should, uh, come back to this. Yeah. I was just going to say real quick again, I think, Bobby, I think what you're articulating, I mean, having read your book and, you know, talked with you a lot about your work, I think we're very much on the same page. Uh, I was very much also at the same time doing a devil's advocate thing though, for a couple of reasons. And it would, it would derail us too much from the present topic, which is more important than trying to make a devil's advocate, uh, case for living with determinism. So yeah, well, I, I told Daniel that I'm a recovering determinist. That's kind of, that's kind of what I want to do here, because I said, if you don't, if we don't hear the criticism, then the people who don't agree with this, I'm not going to answer. So I, I'm glad you said that, but I would just like to push you a little bit more to explain. Yeah. If, if it's purely deterministic, what's, where's the responsibility that Nietzsche's talking about?

Yeah. Well, let me say maybe a couple of things too, um, uh, that, that add further context here. Um, there's sort of a, a tradition, uh, of let's say using historical, uh, philosophical, or even theological categories, um, and the kind of, uh, frameworks that those used to provide for making sense of really important, um, concepts and ways of being in the world. And then presuming that if someone deviated from those, that there'd be these natural, natural, uh, necessary sort of implications that would flow from thinking differently about something. Uh, the classic case in my experience has been, oh, we need morality and moral facts to make us good people. As soon as you don't believe in moral facts, you'll become, uh, you know, uh, a terrible human being.

And that's not true. Uh, you know, people who don't believe in moral facts in the kind of traditional way, uh, can find all sorts of groundings, uh, for their morality.

Yeah. But sorry to interrupt, but I do, I can see like these language things going on, like everywhere in these discussions. So for example, um, yeah, so you're saying that people don't need morality and that these people, but then you said traditional morality. So that was the point I was going to make is that the people who let's say you don't believe that there's any purposefulness in nature, there's no teleology. So it's like, everything's random and accidental.

Then they go, well, you know what? That doesn't make me depressed. I'm actually, this is the only time I'm, you know, this is all we've got. So I better make it count. There's no afterlife.

Um, so what I'm saying is they're still deriving morals from this, uh, paradigm and they're actually deriving meaning and purposeful principles out of this. So they're really, um, um, in a way there, they still see some, they're, they're still finding what they need to find to, um, behave in an optimal way. But what I'm saying is that that's kind of a teleological view and they're not fully sticking to their paradigm with no assumptions. They're building on lots of assumptions and you get this morality that ends up looking exactly like religion. Once you map it out and you go, oh yeah. Like the golden rule, for example, the golden rule, you don't need a religion or the golden rule. You just need to know that you're an agent. There are other agents like you, they're like you. They also experience happiness and suffering. Therefore I'm going to treat them like I would want to be treated. And also there's an incentive to do that because then they can treat me that way. Um, then they'll treat me as I've treated them, but the golden rule could be a basis for a religion and a moral postulate. So are they really moral morally free, free of moral facts or are they creating a set of morals from a paradigm that's supposed to be completely moral free, but maybe it's not because what I'm saying basically is they're kind of converging on this kind of purposeful narrative without even realizing it.

Well, yeah, I mean, there's a, there's a lot there. I mean, I would say what you're getting at, which is important, which is there is often a performative contradiction that we're engaged in.

We're asserting something logically and then our, our actions are sort of, you know, suggesting otherwise. I mean, if I, if I'm a solipsist and I say, you know, I believe in solipsism and I think this is right. It's sort of like, well, wait, why are you telling me that? You know, like, uh, so there's, there's that element to it. Uh, there are actually though, some rather detailed arguments you could make that get into things like, um, well, yes, I'm a solipsist, but I'm talking to you about it because your projection of me, and it's still worthwhile for me to have projections with my, you know, conversational projections with myself, et cetera. And there are versions of that for the free will sort of a thing, right? That's you could, but I think there's fallacies with all of that.

So for example, let's say, and there's a, uh, uh, old joke about the teacher gives a, uh, lecture on solipsism and why solipsism can't be refuted. And the student comes up to him after the lecture and says, Oh, you're a solipsist too. Like he's happy to meet another solipsist, but the whole point is that the other person doesn't exist. So you're happy to meet you. But so what I would say is let's follow that anytime. Okay. So then we would do a little game where we break that down. And what I'm saying, when you start to break all those things down, they lead you back to the same road. So let's say you're not real, Brendan, you're a projection of my mind, but what does that mean? Brendan is a projection of my mind, but you're an agent and you have different beliefs than me, as I know myself here. And you're going to do things. You're going to behave in ways that are different than I would, then I would behave because you have a different set of beliefs ultimately compared to actually the me. I know, even if you're really me, what I'm saying is that really me doesn't really have any meaning because when you start to define you as this agent, even if you make this metaphysical claim that you're a projection of me for all practical purposes, you are an agent within this system that has a different set of beliefs than what I know that behaves differently. So what I'm saying is even the thought experiment has this language fall because you come back to the equivalence of it's not really a meaningful statement to say whether it's me or not. If it has all these properties that counts as an individual agent in the reality that we can be aware of, if that makes sense. Yeah. And I think we, I agree. If we really followed this through, I think we would converge to the same point, which for me tends to be, well, I'll frame this a little bit by saying something like, you know, you know, Hume, you're just mentioning earlier about how, like, you know, Sam Harris said, I wouldn't tell it to my daughter, you know, Hume skeptic would question, well, will the sun actually rise, et cetera. But he made the point that basically once he puts his philosopher pen down and he goes and leaves his study, he's now living like a regular person. Now you could say, so there's a, there's a potentially meaningful distinction between the pragmatism and the sort of intellectual logical, you know, coherence, let's say. But I do think that there's a meaningful way that we can actually collapse those in a way that should be collapsed because, because it would lead to something like what information about reality has a, has causal power for sustaining my viability and flourishing in the face of entropy, right? Kind of tie back in some of this free energy principle, you know, Bayesian updating sort of thing here, right? So, you know, and this is the point that, that you introduced me about the work of like Kolchinsky and Walpert, which I've made great use of, of thinking about meaningful information as being that meaning that a system produces by processing information about its environment that ultimately aids its viability, right? Now, if what we're talking about in a pragmatic sense has a deleterious effect on our viability because of our concepts, then we might say that that actually is decreasing our viability. And therefore we might be inclined to say, if that's our sort of criterion for asserting information as being an accurate model of reality, which I'm not sure of a better one, then we could say that things like skepticism and solipsism are, are untrue on that ground. If that, if that line makes any sense, you could really start to unpack that. But I think that that might be where we'd ultimately converge in agreement on some of this points on some of these points. Yeah. So five minutes. I'm after running five minutes, unfortunately. So I'll just like throw in a quick, a couple of quick thoughts.

David Gardner Oh, let me just say one sentence. Before I forget, maybe I already forgot. Let's see. Okay, go ahead, Adam. I'll try to remember it.

Adam Leffler Sorry.
David Gardner Oh, sorry. Yeah. When you were talking about viability,
like all these explanations like, oh, well, maybe I don't have free will, but being optimistic has some positive effect. What I'm trying to say is if the micro scale determinism is true,
and we're going to break. That's one point I want to make. There's determinism. When people are saying that word, it's conflating two different things. There's physical determinism that I'm talking about,
like micro determinism, where everything is just this model, like where you have these atoms are basically pool balls and they're following these trajectories. Well, I'll get to that in a slide,
but that's a different kind of determinism than saying that your genes and experiences
David Gardner Give rise to what you do. My point is, if you're saying all these,
if the micro scale determinism is true, there's no need to even talk about, oh, optimism helps me.
David Gardner There's no like, oh, even being optimistic. All of that biological level talk
is causally redundant. So basically, if there's micro scale determinism, genes even aren't determining anything, that's causal over determinism. You're saying, okay, the genes cause my behavior,
but then you said the atoms cause my behavior. David Gardner So my point is, if micro scale determinism is true, you don't even have genes determining your behavior. It's an incoherent argument. Sam Harris is conflating these two types of determinism and flipping back and forth.
So what I'm saying is, we got to break this down. What top down causation does is it overcomes this physical determinism. And then there's biological determinism that determines what agents will do.
And I'm saying that consciousness and self-awareness, reflective thought is what overrides biological determinism. And that's when basically your higher mind, like this monitoring kind of prefrontal cortex, executive control, like the mind that's responsible for that. There's access consciousness versus phenomenal consciousness distinction. So that mind can override your biological programming, your impulsive actions. So when we have this hierarchical model, we see that there's multiple, there's two types of determinism at least. And then there's multiple types of top-down causation, which overcome those things. So not to say that that argument doesn't hold up, like we have to talk about determinism as far as like your genes determining your behavior. But my point is,
if someone believes in micro scale determinism, then these higher level things, when people say, oh, you know, your genes are deciding what you're doing, that would, those things would also be epiphenomenal. Those would be shorthands, because you could actually look at the genes, and you could see a bunch of particles, and then you could use Newtonian trajectories just applied to the particles.
The genes are epiphenomenal. Now, the neurons are epiphenomenal. So it's a level of new, I don't know, Adam, have you thought about that at all? Adam Lundberg Yeah, the, um, and that's kind of my,
not part of my agnosticism is like, is there such a thing as a true atomism? Does causation even make sense if it's just like an undifferentiated, like, uh, quantum field, for instance? Like,
as soon as we're talking about particles or kinds of any, anything we can even name or pick out,
um, are we basically core screening? And are we in kind of like a, um, and, and then like,
when we say micro, uh, why don't we just keep going? Like, isn't it already somewhat like an

instrumental for a given purpose? But in terms of like, so like, I've been like, and I've kind of kept these separate, like these issues of like, the philosophy of causation, like the handlings of free will. So like, I've tended to focus on like, functional cybernetic free will, basically, as a the planning and, and like the, the, the architecture of choice and like being able to like, um, suss out different options and evaluate counterfactuals that you then like, uh, have different control energy for the system. But I think there could be really interesting connections between how we handle causation though, and where within the control hierarchy and the way you're describing of like, we're, um, what's responsible for what, so like ways in which like system two can override system one or like, that's absolutely what I want to develop here, because I think you can bridge this low level causation discussion that I'm talking about, which for example, Paul Davies and Sarah Walker have written about informational control and that's with the origin of life. There's a change in the causal structure of the system and then it gains this agency. And that's what I focused on a lot in part one of the book. Um, and so what I'm saying is Adam's work focuses on basically this free will topic where I'm saying that there's this even higher level system beyond just normal biological agents that comes with not just consciousness, but an ability to create counterfactuals and imagination. So it's like a prefrontal cortex sort of higher self. Um, and a lot of people say the self's illusory. That's why I love to fall back on Adam's work because it really breaks down the self from a mechanistic perspective. Um, but that's exactly what we can do. We can look at the space of counterfactuals. And as we will see that I think there is some indeterminism at the fundamental level of reality that propagates up. And that's what, you know, you see with chaos, even though it's called deterministic chaos, and that's a whole nother issue to get into. But basically, um, I'm saying that there are choices and that the world's trajectory is not set in stone and there's some variation. And when we as agents, um, consider the space of counterfactuals and then choose something specifically, um, it may be something like we're going to make a choice and then our biases, like our impulsive decision is to make a certain choice, like reflexively based on like how we feel and, you know, our genetic programming and our experiences, but then we can have another process where we go, okay, is this decision optimal given my principles? I have this higher self. Now I think about like, what's good, what I want to do, what are my long-term goals? And that, that is the mechanism that corresponds to what people mean when they talk about free will. Um, and so I'm excited about that, you know, and, and, and working with Adam to, to explain that to people who, you know, are just saying that there's no way free will can work. Cause he's looked at the, talked about these mechanisms from this, you know, cybernetic and computational neuroscience perspective and formalize them. People like Sabina Hossenfelder, like physicists are like saying, there's no way for you can, free will can be compatible with a causal picture of the world. I don't think that's true, but what I wanted, you know, and Adam's going to leave. So he's unfortunately not going to be part of this, but he'll see it later. I think we can bridge Adam's work at this higher level to this low level causation stuff to actually have a causal story where there's actually still cause and effect, but there's higher level causation. Um, that can basically override the sort of cause the lower level bottom up causation. And then you even get these higher levels where you can have, um, um, that sort of causation being overridden by conscious, like mental causation. So you would have bottom up causation and then top down causation and then mental causation and top down and mental causation. They would all be like looped under like, you know, categorized in or downward causation, or you could call them all top down causation, but there are differences when consciousness emerges that allows for what could be called new causal power. Things are working in a similar way, but you will see a behavioral change when you get consciousness that will allow for things like technology, things that require imagination, uh, counterfactual generation that can't be achieved with the lower level intelligence. And that we can actually construct a causal picture that is perfectly in line with everything we know about causality. But when you do this, you realize that the bottom up story, and I haven't really defined that, you know, you guys know what I'm talking about, but I haven't defined it for the listeners, but that when you have this story of causation, you have agents emerge and they're actually control centers that can initiate causal chains. And they do it for reasons. There's always reasons that you want to make a certain decision over another. So some people think because you have a reason, that's a cause that somehow that's not free will. But if you didn't have a reason for it, you would be doing it randomly. And that wouldn't be free will either. So free will just has to be reconceptualized within this framework. And it's not full free will, but full free will wouldn't be free will in the, in the first place, because again, it would be random if it was not based on a reason. But what I'm saying is that your reason, even though it's a cause, the reasons it's not determined, because basically, it's, it's not determined in the same sense that people are talking about with physical determinism. We'll get to this, there's no way I can do it without looking at the slides. But if anyone has any comments about what I just said, and Adam, I know you got to go. So any final concluding thoughts? I am furious that I have to go. But I'm really, I'm looking forward to tuning in a bit. So talk to you soon.

All right. See, Adam, thanks for coming.

And yeah, if I may, sorry, may I interject for a second?

Yeah, if anybody wants to read Adam's papers on integrated world modeling theory and cybernetic free will, you can just Google his name in those terms, and you'll see some great papers and some, you know, really respected journals like entropy, I think is the integrated world modeling theory one.

Yeah. Michael?

Michael Leach

No, I was just, I'm just trying to, again, do my job here. But Daniel, I did want to ask your thoughts about, you know, active interference or inference at this juncture.

Daniel Schroeder

Yeah, any thoughts about active inference, how that applies to anything we've said already?

And I will link it to that soon, because I know a lot of listeners, that's what you're interested in.

Daniel Schroeder

I'll definitely look forward to how you link it. But although active inference is this incredible, modern, composable framework for on one hand, cognitive modeling, and on the other hand, the actual embodied situational modeling. A lot of the reason why people get into it, get curious, start participating in the Institute is exactly the topic of self-help and self-improvement and personal decision making amidst multi-scale uncertainty. And so there's among the many gaps between theory and application or practice. One of them is utilizing these kinds of multi-scale information geometry, decision making moment of choice type frameworks for what, for what we actually do. So this is something that really strikes at the heart of what we do as inference. And I'm just looking forward to how you frame it in this narrative. And then we can come back to some more technical details too.

Daniel Schroeder

So we're going to cap it at eight, I guess at 7.50. I don't know if you'll have many questions in the chat.

Daniel Schroeder

But I'm going to try to give the presentation in the next 45 minutes.

Daniel Schroeder

But feel free to comment on stuff that will still be helpful.

But I guess we shouldn't get too far into like, make a comment so we can mark it, I might give a little answer and then we'll come back to it. But I prefer that you speak up because then we'll know what to come back to.

Daniel Schroeder

So I'll say a few more things about causality because they're important. And then I'll get into kind of the bigger picture of the theory and then specifically the self-help teleological stance, which is, you know, the free energy principle comes into play there.

Daniel Schroeder

And, um, yeah, then we can talk about practical applications for the individual and for society, because the goal is that this system can be used. It's kind of like a theory of everything. So it explains reality, but that it can be used to improve your life as an individual, but also be used to figure out how to sort of optimize our social systems, political systems. So it kind of creates a logic that could be used to base like a new, like a political party on like a third party or something, which would be great if we had a party kind of based on systems thinking, are we kind of need are not like a bipartisan ideology right now that's not based on anything other than, um, evidence based things that we know can be used to to optimize systems? Not that that that's everything but what i'm saying is it provides a foundation that then the ethical uh and kind of moral stances emerge from and if you don't see this basic cosmic story then you will be confused about ethical issues and what i will argue is that folk psychology ethics all of that stuff it was it's i i believe is consistent with this view because it was kind of intuitive that this is the story of nature and that it will kind of ease that cognitive dissonance for people who have accepted the reductionist view like i know a lot of people have accepted this view that there's no agency that there's just the strict determinism but then they want to use you know this free energy

principle type things to improve their life but that's kind of a contradiction because
are they choosing to improve it or is there just one trajectory that's set in stone and and and so i
think that getting this stuff straight will allow people to embrace stuff like the free energy
principle and apply it to their lives because they go okay this i can actually use this and i can do
that and i can create change in myself and in the society i'm a part of but with this old model
there's no hope for change because there's only one trajectory and so if nietzsche really took that
seriously it would be no reason to really talk about responsibility because it's just purely a block
time frozen thing where the future's already determined from the initial state and the fact
that it would repeat over and over is just even more hell it's like determinism was bad enough now you
have to experience not having you know an illusion uh forever um so what i want to offer is a different
model that we can't fully get into but what i what i have called eternal recursive emergence and so that
it's not a cyclical process it's actually a process of the universe evolving and maybe giving rise um to
something new potentially like a new universe um there's cosmological natural selection we don't have to go
there we have to talk about like what what you know anything outside of this universe but um
my point is that um yeah so what i want to show is that the universe is uh not an arbitrary system
it's not random it is a system that exhibits a design you can say it's the appearance of a design
maybe it's not really designed but there's there's qualities that can't be explained
um if the universe was the only universe in all of reality uh you you couldn't have that theory
either there's a designer that designed it that way or there's a larger reality with a multiverse
that would explain this fine tuning and that's where basically science went they said okay it looks
like there's a creator because the fine tuning problem but we can't accept that so we have to consider this
multiverse option but the multiverse option is just as weird as a creator it leads to all kinds of crazy things
like other universes where there's every possible version of you um so it probably has
this just as much baggage as the creator thing um but then there's cosmological natural selection
which basically is an evolutionary explanation where basically universes give rise to other universes and there's this
and there's this darwinian selection principle that would lead to like a universe optimized for like complexity growth
and life uh essentially like spreading like continuously so we don't really need to get in those metaphysical topics but
uh i just wanted to state up front that um the universe does have something that uh we could call a design
but when i say that don't assume that i am talking about it being created by some sort of conscious intelligent agent
it could be this evolutionary process um but it's not random it's not arbitrary and that's the big mistake is that
we have models of a random system like a gas that's what boltzman used to kind of
try to have a statistical foundation of the second law of thermodynamics and the problem is we base the dynamics of that inanimate system
i mean we we understood the dynamics of that inanimate system and then we assumed that the universe behaved the same way

and that led to all of these um bad assumptions that created this nihilistic world view so let me let me talk about that first
if determinism is true like if there were no indeterminism at the lowest level for example um
everything was just made of these atoms and these atoms were like
classical scale objects like pool balls on a pool table and that you have all these atoms and if
those atoms follow classical trajectories and they're just like basically you can imagine the universe as a
table of pool balls um then there's no room for anything there's one future trajectory
and it's set in stone so if you believe that if you accept that worldview there is no other future
there's no changing it and if you believe that and you want to change your life you have to and
you're really taking this seriously you would have to ask yourself well if i change my life am i really
changing it or was this the way it was always supposed to happen it was determined that i was going to change
my life or are you saying like it's determined in a way where you can't change your life so you shouldn't
even try um it leads to all these weird conclusions but what i'm saying is that's not the way reality is
and that leads to like this worldview where you have no agency you can't get around it because if
it's a purely pool ball universe then there's no agency um not only that so that's the first thing
that's if if the world was newtonian at the micro level there would be no agency there would be one
trajectory and there would be this block time thing it would just everything would follow from the
initial state in a way that was calculable from just specifying the positions of every particle and
applying newton's laws to those and then we have some forces that interact but it doesn't really
change the picture um so there's just one continual trajectory then bolt's been found that you know
an inanimate system because all of in a gas for example so it's a specific type of system that
boltzmann was using to derive the second law of thermodynamics on a statistical foundation um and
it's not the way open systems work so first of all we should know that um he showed with statistical
mechanics that if you have a gas so in a gas every particle is uncorrelated it's just moving around
freely and in his model also people should know that this is before we understood the periodic table
periodic table didn't come to like decades after boltzmann did this work so we didn't understand like
chemical bonding for example we didn't understand a lot of things so in his model as an idealization you use
spirit spherical like what would be marbles like spirit spherical round things that bump off each other
elastically so they just bounce off each other they don't combine and interact with forces and if you
have this picture if you if you have these things just bumping off each other interacting this way moving
around um following these classical trajectories um not bound in any way you will get a transition from order
um to disorder if the system starts out in an ordered state of course
because the disordered state there's more possible ways to do that so if everything's moving around randomly
you see this trajectory here that we see um and then so the idea but there's so much there were so
many weird assumptions in trying to apply this to our universe like when you start to break them down
like none of them really made sense so he's saying okay this is what has to happen you go from order to

disorder um and this must wait be the way the universe is because this is what has to happen by statistical laws if everything's moving randomly if there's no magical forces doing anything this is what happens
um so um the problem with that uh basically
well so one thing i should say is that's not even true if this was the way the world is um
um and people pointed this out to boltzman given enough time if these are following these mechanical trajectories even the disordered state will go back to the ordered state given enough time this is poincare's uh um recurrence theorem um kind of similar to nietzsche's eternal recurrence but so the idea is that if you have the system behaving this way you it would um basically all trajectories are possible
and given enough time it will go from disorder and then will revert and people challenged bolstman with this and he couldn't get away from that that possibility um what i i lost my train of thought
earlier um i didn't let you know i just went on to the next thing but what what i forgot to say was that um bolstman was like faced with the problem of why is our universe looks so complex
if it goes from order to disorder and then that's where like so he had to say oh well it must have started out really ordered because it can't go from disorder to order so it must have been this highly ordered state which was the beginning of time it's this low low entropy state but now we know that after the big bang everything was homogeneous and then things started aggregating so it was literally an equilibrium state this fully mixed up high disorder state something closer to that
before that and then things started clumping and you get more organization so the story is completely the opposite of what we've been taught julian barber in his new book explains this that the trajectory of the universe is towards higher complexity and order and that we made a mistake by applying this model of an inanimate system to the universe in its entirety when the universe has all of these batteries in the sky we call stars and they're driving systems away from equilibrium so then we had um
ilia prego gene in 1940 1950 1960 1970 i would say 50s and 60s were probably when he was starting this work
and it was really gaining influence but he basically showed that there's another dynamic where if you have a system of particles driven by a flow of energy it will start to naturally organize that system um assuming that the molecules can bind to each other so you need to have the right sort of um molecular variety for this to happen but this model of boltzmann's didn't take into account bonding it didn't take into account the force of gravity it didn't take account into account open systems and what i'm arguing is ultimately that the universe's dynamics are more like an open system being pushed far from equilibrium where it's self-organizing so in this model the universe isn't going towards disorder because it's not a
a universe made of particles that that stay uncorrelated so this bolson's model is based on this h theorem where everything stays uncorrelated but when things interact they become correlated so this model works for gases and it doesn't work for open systems and what i'm saying is it doesn't work for the universe we applied it to the universe and if this was true there's only one trajectory which means
there's no changing the future what's going to happen is going to happen and if that's the case
that's pretty depressing because it kind of demotivates us to go oh you know what i can change the world i can have this idea and it can blow up like mark zuckerberg by creating facebook like you could create

something like facebook and you would have that and you would change the world but if you believed in determinism it would be kind of strange you'd be like wait but this is already determined so i already did it in a way i'm already going to do it and like there's only one trajectory and then i play that important role in history no that can't be true i'm not gonna that's just impossible like i'm gonna change the world and it was determined so that philosophy leads to a lot of weirdness that i'm saying completely dissipates when we get the causal picture straight and we see that there's not this nihilistic one trajectory because of quantum indeterminism and that's not everything but what quantum indeterminism does is it shows that when you shrink when you zoom in and you get to a certain level that newton's laws of motion do not hold you will see there will be no definite state of an electron that you can predict given its prior state there's only going to be a wave function that gives a distribution of probabilities where that can go and then use the born rule to find it where it's likely to go and then if you repeat that study a bunch of times you'll see these statistical patterns but so what i'm saying is quantum indeterminism get rid of this picture of strict determinism at the micro level and if you did have that strict determinism at the micro level you would not have any agency or any there would be no real counterfactuals they would be sort of due to our epistemic limitations what we can know about the world we think there are different possibilities but there actually are none the world can only go one way and what i'm saying is that there are some different there's a possibility space of things that can happen and that when agents make decisions you're carving out a path um that in some way is not it's not determined by the micro scale physics because of quantum indeterminism and then we have chaos too at the classical scale where you can't predict what a chaotic system is going to do people call it deterministic chaos i have issues with calling that determinism because it's completely not predictable and i don't even think determinism is the right word people want to look up norton's dome if you have a ball posed at the top of the dome there's multiple newton's equations that can satisfy where it's going to go next there's actually even determinism is not fully deterministic but that's kind of beyond what we'll get into so um let me just try to go faster through the slides since the 720 so i'm not going to talk about a lot of the details that uh i planned but maybe our uh earlier uh conversation provides context that can clear all this stuff up so what i'm arguing is that the universe is self-organizing and so lee smolin uh popular physicist says the possibility of conceiving the universe as a whole as a self-organized system in which a variety of improbable structures exist permanently without the need of a pilot or other external agent um i can't see that word um so we have the possibility of constructing a new scientific cosmology um it's clear that if the natural state of matter is chaos and external intelligence isn't needed to explain the order and beauty of the world um so he's basically saying if we had this model of like randomness then it wouldn't make sense for us to see what we see and we would need a creator would be a better explain explanation but if life order and structure are the natural state of the cosmos itself in our existence indeed our spirit might finally be comprehended as created naturally by the world rather than unnaturally in an opposition to it and you'll see up here that he says a variety of improbable structures um that exists permanently so what i'm saying is there's some language issues here that we're seeing um where we're saying why do we see all this improbable stuff if it's there it's not improbable so it's only improbable if we accept the old model uh based on you know this simplified model of causation that doesn't include top-down causation which we'll explain and that actually complexity that you would say that this is highly improbable that complexity just keeps growing and growing and growing because statistically you would think that there would be some transient complexity and then basically you would have this trend towards entropy that you know this complexity would dissolve what i'm saying is it's not when i'm saying that the universe is becoming more and more complex it's not opposed to naturalism it's that our conception of naturalism is based on an outdated paradigm that did not include dynamics that really exists that we're now just finding so that this these structures that are highly complex were never improbable and i'm actually going to argue that they're statistically inevitable um uh that they arise um in a way deterministically and it might be kind of confusing that i use that word but what i'm going to try to argue is that there when we talk about determinism we need to make distinctions between that as well so what i'm saying is that there's strict determinism is not true but i'm now going to argue that something like global determinism is true which is a name that's been used by philosophers that there is that there are attractors that create these dynamics such that it is natural that the world goes towards increasing complexity and that if we see this trajectory that we're a part of you know life is spreading we're building technology the biosphere has created all this complexity on earth if we spread we're spreading complexity throughout the universe in the form of like you know computational structures life is a computational structure this information processing network that we call the biosphere is spreading if life spreads that that was never improbable it was just improbable because it seemed improbable because we didn't have all the laws and that actually it's inevitable um but that there is some freedom within that but there is an outline that's basically uh deterministic at this larger scale um and by that i mean things can go a different way like if you were around the tape of time and you played it again that different events there would be a slightly different chain of events that would happen you would get different for example genetic mutations because genetic mutation and schrodinger uh was the first person to speculate about this but that there's a quantum component so if you rewound the tape of time and you played it again you would get some different genes but what i'm arguing is because of these emergent attractor dynamics that were really explained by non-equilibrium physics and in the work of the pre-gagene then what you see is that you would get the same trajectory towards higher and higher complexity because there were other laws that we weren't aware of and so in this model there's going to be a sort of cosmic destiny that is this sort of form of global determinism and then there's going to be free agency uh that the agents can do things that's not set in stone within that kind of global deterministic picture so it's kind of new it's not i'm saying that's not binary is the world deterministic or indeterministic there's indeterminism at the lowest level which basically introduces slack into the causal chain so not everything is fully determined in the strict sense but because of these attractors you get these emergent dynamics where you're going to see similar patterns similar regularities um so christoph coke just a couple scientists who well yeah i'll skip this stuff but so there are a lot of scientists who a lot of philosophers who have had this kind of teleological narrative that the universe was moving towards some sort of goal that it seemed to be like an organism that's developing and that seemed like a an idea opposed to naturalism people like thielhard de chardin and uh bergson were you know um uh in chardin was religious but bergson was a philosopher that wasn't really so i think a lot of these people got criticized wrongly they're actually seeing what was right about nature but this view was thought to be opposed to naturalism that was teleological and that had to be some sort of mystical force and what i'm saying is we just didn't know about certain things we didn't understand information and that these people were right and that a lot of people will hear this and say oh well that's against science that might be again they might think it's against the second law of thermodynamics or something let's know there's a big misunderstanding about that and all of these people are you know all the scientists on this page really everyone with the exception of a couple of those philosophers are extremely uh respected and very accomplished people who support this view i mean ray kerswell you know he's technology guy it's a little bit more out there but he even uses the word destiny destiny ultimately it is clear that the physical laws of our universe are precisely what they need to be to allow for the evolution of increasing levels of order and complexity ultimately the entire universe will become saturated with intelligence this is the destiny of the universe so what i'm saying is that this is the trajectory of cosmic evolution this is the the roadmap of the future and this provides something like an outline for a non-reductionist theory of everything based on emergence so you have first physics and chemistry emerged in biology then brains and technology and then the merger of biology with technology and then you get patterns of matter and energy and the universe becomes saturated with intelligent processes and knowledge so life expands and we do that because we have to we're going to run out of energy on this planet our star is going to die which necessitates that we move outward and then we extract the energy from that and we keep moving outward and as we do the informational network that is life keeps expanding and it keeps taking inanimate matter in the world and incorporating that into its computational substrate so the the island of agency the island of information keeps spreading and subsuming the inanimate world and i'm saying that that is an inevitable process and people say well that seems very unlikely and if that that's happening then it's it's not part of the the designer it's not part of the the the the code of the universe like it's not necessitated by the the parameters the laws and the constants they'll say you know it's just something that could have happened uh and it's unlikely but we're defying the odds now when those people say that they don't even realize their own ontology like if they believe in determinism then there's only one trajectory there's only one thing that could happen so there's no likely or not likely if you really believe in determinism there's no statistics and there's really no causation it's just one thing follows another and people have talked about that if determinism true then even causality is an illusion it's just one sort of fabric of events that all follow from the initial event so um what i'm arguing is that the universe shows the dynamics of a far from equilibrium system a driven system and that it moving towards complexity isn't violating any laws and it's not improbable we just didn't understand far from equilibrium dynamics and that the universe's

complexity doesn't have to be transient so sean carroll has this thing where he's showing like the increase in complexity and the increase in entropy and people go okay this is making sense complexity is not opposed to entropy it's part of that but it goes up and then it goes down you have the coffee and you have the cream and the the coffee mixing and you get this complexity and then it all becomes homogenous what i'm saying is no that's not true for the universe at all complexity keeps going because when life emerges it's not like some complexity like you get some you know random pebbles arranged as a crystal and then the wind blows and then they disperse life is robust it's modeled the world and it's because it's modeled the world it can start to it can maintain its own existence so for example if an asteroid is coming at the earth in 100 years assuming that we advance technologically at the rate we've been we're going to destroy that asteroid we're going to do what we need to do to preserve life and that's going to change the deterministic trajectory of the inanimate world so life changes the in the trajectory of the inanimate world but what i'm saying what it was inevitable that life emerges and does this so there is a sort of trajectory that's sort of set in stone from the uh initial parameters but that this trajectory um is the opposite trajectory of what boltzman thought it's actually towards greater and greater organization and the discussion of where's the limit if this universe wakes up and the whole thing becomes this integrated computational system when does it have to stop what about the second law of thermodynamics what about the energy uh needs and i wrote an article for noema called life need not ever end and it was shared by david deutch and sabine hausenfelder and robin hanson also were commenting
on it and saying yes it's possible that life could potentially not have a limit depends on if there's an upper bound entropy robin hanson said sabine was like i have a chapter on this in my book so a lot of the things that we took as gospel like the first law and second law of thermodynamics those apply very well to certain systems but we shouldn't have applied them to the universe as a whole i mean the first law of thermodynamics is just like uh there's conservation of energy which which is um holds and we use that to calculate things but this statement matter can't be created or destroyed absolutely not true cosmic inflation theory has new matter and energy created during the cosmic inflation period um continuously and then now with the expansion of the universe there's dark energy constantly being created as well and astronomers i mean there's this astronomy conference paper where people have speculated have argued that we can use dark energy and that's getting more speculative um uh to continue to to power life um we might not even need that if you read my article you'll see this but you can look up on sean carroll's blog he has an article called energy is not conserved and it talks about basically the first law is not true in the way some people thought it was um whoops didn't mean to do that let me go back to this slide so um let's speed it up um so maybe this is where we're at now so it's a series of these um stages and what i'm saying is we're at this stage and that the biosphere um mirrors a brain it's an information network of agents exchanging information through the internet and that there's actually you know people might say that's cosmic mysticism or there's some interesting new theories by people like lee smallen and drawn linear um arguing that the universe is a learning machine and everything in that paper is consistent with this view so go read that paper read the universe's neural network by vitali he has

a group on facebook and we've been meeting um we're taking a break for the summer and making some videos but basically there's a mathematical foundation for this this picture and uh so calling it cosmic mysticism it's just not true anymore and you can have a fundamentally purposeful universe

headed towards this what tlard de chardin would describe as like an omega point um and it's

completely consistent with naturalism as long as you update naturalism to include these new dynamics

so um i was going to talk about phase transitions so basically this these stages in cosmic evolution occur from components like nature's fundamental particles combining to make higher level systems so we have you know atoms uh come together to form molecules which come together to form cells which come together to form organisms and ecosystems and in the biosphere is this fully integrated network so those are phase transitions it occurs in stages you have a depiction here we see a single-celled organism

uh and then those organisms you can think of a slime mold so you have a bunch of like paramecium or something and they need to solve a problem and so they come together and they solve a problem they form uh multi-cellular organism so here's a slime mold here's a human and these agents also come together

to form larger collectives and every time you have one of these it's called a phase transition and it's

exactly the same as the phase transitions not exactly the same but the same category of phenomena like when you have

um water freezing to ice but we have these non-equilibrium phase transitions where you'll have like the emergence of a a tornado or a whirlpool in your bathtub um and basically these little units like in the formation of a whirlpool you have all these atoms that are bouncing around chaotically and due to some thermodynamic constraints that we won't get into here they start to organize themselves into these uh collective formations that create these flow patterns and you can see these organized structures and

that life is basically another version of this um and that when um atoms come together to make uh

or subparticles come together to make an atom like these phase transitions they continue in the hierarchy of life so you have agents coming together single settlements to multi-cellular then to make societies

and these phase transitions um happen at this edge of chaos so there's this sort of dialectical dynamic between order and chaos that creates complexity and really everything is going through these cycles so when there's a phase transition you know you don't want a system to be too stable because it's too rigid

to transition to the next stage so there will be periods of chaos like we're experiencing right now in society and you might think that the world's ending but it's actually an indication that a phase transition is about to happen and so the system has the potential to either collapse or to transition into

a higher state of organization and so you know this shows this these different stages you have non-living then you have it with living you go from cells to multi-cellular organisms to tribes and cities nations and then to gaia and then it can even extend from like when we terraform mars that's the biosphere self-replicating

and so life will basically the spread of life is like this higher level unit that's the biosphere

making copies of itself with variation and selection and then you know life would potentially you know um

um i'm arguing inevitably subsume um galaxy uh solar systems then galaxies uh and and so on so

let's skip some stuff and get into the the self-help stuff since we don't have much time

daniel are is it at eight do we have to cut it eight or can we go ten minutes longer
we can go ten minutes over well we'll only go ten um i don't want to keep you too long but that will
help with the questions if i can end at eight then we can do ten minutes of questions sorry maybe i'll
try it in before but um so these phase transitions create these memory systems so these higher organized
systems um like life uh so when we get the first cell we have genetic memory and that's encoding
information that the organism basically that living things have discovered through evolution so evolution
is this process where it's making these different copies of these organisms um so there's a genome
with some information but it makes a copy and that copy has some variation so you get something different
and that difference that organism might be better fit to the environment or worse fit if it's worse
if it's a worse fit it's going to die out so it's going to be weeded out that that template of information
is going to be weeded out of the population and the the the organisms that are well adapted
get to make more copies of themselves so what's going on in this process is that you're building up
information in an organism and that information corresponds to adaptations and so that information
is information that is predictive about the environment so organisms are performing inference from the very
beginning of life's emergence they're trying to solve a problem it's how to stop this order to
disorder transition basically for an organism to stay organized it has to find energy and if it has to
find and that's what schrodinger's book what his life was showing is that you know life resists this
tendency to fall apart that boltzman described because it captures energy and that energy allows the
system to do work that's keeping the organization so with these phase transitions you get different
memory systems building on each other so you have genetic memory of the origin of life then you have
neural memory with the emergence of brains and then you have the emergence of civilization you get cultural
memory and then you get meme propagation so you get a different mode of transferring information and
all of these memory systems build on each other and they all encode predictive information about the
environment that allows the living system to stay out of thermodynamic equilibrium to stay out to
prevent falling apart um so one thing that is worth mentioning is this new picture of hierarchical emergence
this new paradigm where you have this multi-level level nature of reality where reality is organizing itself
into these hierarchical levels it has predictive power that the reductionist theory any reductionist theories of
everything don't have those are micro physics theories and they don't apply to life so those theories of
everything i would argue aren't theories of everything because they don't include life
and civilization and consciousness and what i'm saying is that that stuff has causal power it's not epiphenomena
so those things any theory of everything needs to include those things because they consider them
epiphenomena they go we don't have to worry about it but what i'm saying is those things actually have
causal power they need to be explained so some predictive power that this theory has that the old model doesn't the
micro scale model is that we can actually use the dynamics of a system at one level to understand the
dynamics of the system at another level because they're isomorphic they're they have similar patterns
because they're all complex adaptive systems so what i'm saying is that you can make predictions and you

can
use principles to optimize society that are the same principles at work and evolution that optimize organisms and so we can think about someone's asking okay maybe you accept that there's a global brain you go okay we are agents we're informationally connected we're exchanging information the way neurons exchange uh you know signal each other through action potentials we must be forming this collective intelligence the way that there's an intelligence in the brain and we know we are because that's how we build rockets and stuff the single person doesn't do it we build on accumulated knowledge we stand on the shoulder of giants and we share information that's what science is all about it depends on that so if you want to ask what how intelligent is the global brain is the network of humans then you can fall back on understanding the way that neuroscientists have understand brains and what leads to greater computational power and brains or is what's the most robust design for a civilization well what's the most robust design for an organism well we want it to be hierarchical because you have this division of labor and all these other advantages that we'll talk about in a minute but so my point is this theory allows us to predict things that we couldn't predict before because we are using a model of a similar system of a complex adaptive system at one level to tell us things about what hasn't happened in the future and what will emerge and so there is a an essence there's a aspect of determinism here it's not the same kind of determinism but there's a global determinism where you get expected patterns at these levels so we can get a glimpse into where the future is headed but we can never know the details and we can never know what will emerge exactly but we can see that these phase transitions happen and we will get new emergencies um talking about predictive power so paul krugman nobel prize-winning economist said by 2005 it will become clear that the internet's impact on the economy has been no greater than the fax machines probably the most wrong statement that you could ever make um and so um why did he make this mistake here we see it again internet may be just a passing fad as millions give up on it why did they make this mistake what i'm saying is if they had this framework that i've just outlined we would have known when we saw the internet this is a tool for interconnection this is a tool for complexification it's not going away maybe the internet would go away but something else with a different name would replace it because this is part of this trajectory that ray kerswell called like the destiny for life so without this theory we will not see obvious things that would be obvious to us otherwise and we'll make terrible predictions like paul krugman and hopefully everybody reminds him every day of his life that he made that prediction um he is into he's really into systems thinking and so i think he would understand this and i'm interested to know whether he got into systems thinking after he made that prediction or before because if you see this picture like tealhardy chardin's newest fear he predicted this the internet like you know 100 years ago um based on his theory becoming more integrated um so what does this mean for society we'll mention some of that stuff before getting into the bayesian stuff just because it's here isolationism is bad we want to be connected if you cut people off it's not good um there's no information exchange and it could be good for your nation in the short term but eventually what's happening elsewhere in the world is going to affect you because it's one integrated system so we want to connect to people because we want to open up

channels of information exchange but we also want to uh preserve nations we want to have this hierarchical thing where there's um control centers at all these levels towns cities nations uh etc so we don't want it to become a homogenized hive mind why because complexity is in computational power it's a function of variety and integration so something is more complex has more computational power if there's more diversity among the parts and if there's more um connections between those parts so that definition of complexity that lets us know if an organism is more complex like a human brain is more complex than an ant brain because it has more neurons and more connections and more hierarchical levels we can use that to create a more robust society now what does this mean for you it means life's part of a cosmic process with a goal there's this cosmic attractor and we're agents that are part of that bringing that attractor about this process allows the universe to to wake up so life spreading is the inanimate world finding a form that models itself and slowly the the universe starts to become this integrated network that's this coherent intelligence so that's pretty interesting um spiritual implications to that however you want to interpret that but when we're part of this what does it mean for us okay so this is the framing of the free energy principle it says organisms are so saving it for last so it'd be climactic so the free energy principles it frames it with this schrodinger framing and carl fristen that's where you got it from and it's also influenced by cybernetics but if you look at that stuff you have the free energy principle because it basically says all organisms will fall apart all ordered systems so free energy principle applies to aliens any any organized system anywhere in this universe will have to first of all extract energy if they want to stay organized uh the free energy principle doesn't focus on that but even though it has free energy in the name but i will focus on that because i think it should be a part of it organisms have to capture real thermodynamic free energy to stay in the game of existence to capture energy from a chaotic external world you have to start modeling the world and that evolution is building a model of the world through variation and selection through through these accumulation of adaptations and then brains do it even better and then you can update the model in real time with brains because you have synaptic plasticity you don't have it's not slow where things have to die and you have this phylogenetic learning which is like the population is learning this individuals can learn in real time and so you have to model the world and the free energy principle that's what it says it says there's reality and it has all these details that we don't know and we have to create a model or a map of that world that's a coarse-grained simplified version of that and then um that uh model is going to have errors and so what the free energy principle and active inference framework says is that when we're doing actions in the world those are like miniature experiments and um what we should do if we want to live optimally and stay out of thermodynamic equilibrium if we want to survive and be robust we want to decrease the difference between our model of the world um we want to decrease the error of our model and we want to make it more of an accurate representation of the world so you have reality and then you have a model of reality and you want your model to match reality as close as you can to be able to extract energy and avoid threats and to stay in existence so we can be part of this continual process of complexification so we only got 10 more minutes but i think we can uh actually cover all this so this looks really complex but this is going to allow

me to explain the free energy principle and merge it with evolutionary theory and even the process of
science so this is an organism imagine this is the first organism um that uh emerged with the origin of
life and so this organism makes copies of itself and you get all of these different designs right so
these different designs are variations on this model this organism is an embodied theory about how to survive
about how to stay far from equilibrium so it has a predictive model of the world that makes copies of
itself and you get all of these competing predictive models and then these models compete for survival and
some die out and the winning design is the best predictor it gets to make copies of itself and you
get a similar design so what's going on here is you have models that are competing models the way
scientific models compete and bayes theorem is basically you have a theory and you have competing models
and then you make predictions with those models and then you test those predictions and you update the
probability that each model is true based on the results of those experiments and so you want to
update your model based on new evidence and so what i'm saying is evolution when you create all these
organisms these copies and they compete it's competing models the way that scientific theories compete
and the winning model uh is a product of natural selection the same way that the winning scientific
theory is a product of surviving criticism and testing so that's a natural selection pressure
um so as these generations go on you get um with each generation you get an update to the model it's a
bayesian update and so the model starts to match reality better such that after a bunch of generations
you get an optimized organism that's optimized for that niche
and so this is how evolution itself is a process of bayesian inference as uh organisms adapt they're
updating their model carl fristen has understood this equivalence and and says that all of evolution
is inference of course someone might say all inference is evolution and it would all be equally true but
so this model applies to science it applies to learning so instead of all these different organisms you know
which we said could be theories different competing models you could have this as different
models of um maybe you're trying to figure out how to play your sport a sport like to the best of your
ability and you want to test out these different things and then you test them and and one performs better
and so that's the free energy principle is that you're decreasing the difference between your model and actual reality
so you're doing these little experiments all the time and based on what works you're updating your model or
you should be if you want to optimize your system for survival and growth and so um
it's the same when we're trying to find a um uh uh uh a political philosophy
capitalism socialism these are competing models we test those models out this is variation and selection
and the ones that work they get retained by society and we keep those so capitalism i would argue um it's
terrible right now but it was the best model for the time and um the superior model is always going to
be it's never going to be one thing it's always going to change so we would want to test hybrids between
socialism and capitalism and crypto economics if you want to define like determine a political
party based on this what you would say was that the best solution for a political party is to try out
different models in small areas and uh based on what works then you can see where to apply those models
but i would argue that it's not going to be socialism for every place it's not going to be capitalism

it might not even be a hybrid it might even be something in addition to that like i said crypto economics and that we have to test these things and if we're not testing them and we're not trying these different models we're not living optimal we have no way to get uh to the to the to the optimal worldview political philosophy if we're not doing this testing and bayesian updating so that's one big thing uh another thing that you know as i said that you know this is a basis for a self-help system so you have to model the world and your model of the world how accurate is determines how successful you are in life if you don't have an accurate model of the world you don't understand how things work for example let's say you take a bunch of physics courses you can become an engineer and if you don't understand how things work and you're not paying attention to evidence and you just have some sort of beliefs that you believe for no reason and you try to get through life you will not be able to achieve goals and progress uh in the way you would if you were doing bayesian updating um so when i say this is a self-help system basically i'm saying there's lots of inaccuracies about our model of the world that could be optimized if we took a bayesian approach to reality let me see let me tell you what i mean by that in these last few minutes so we have five more minutes so there's this principle of incomplete knowledge that comes from cybernetics and it says our there's this phrase the map is not the territory so our model of the world is not the same as the actual world it can't hold all those details the world is too chaotic uh and there's just too many um there's too many variables for any brain to model so we need to have this low res like coarse grain version of our model of the world and it's going to be incomplete it's always going to have uncertainty so anyone who's certain about anything like sam harris is 100 certain that free will can't exist even though there's lots of uh well-respected people that know a lot more about him than neuroscience and causality that say that free will can't exist um it's just bad to be that certain because anybody who's been that certain in the past has been proven wrong everybody thought newtonian mechanics would explain everything then we had general relativity and now we see general relativity doesn't explain quantum mechanics every model is going to be proven wrong so one thing to do is never be fully certain always take into account the uncertainty of your model what should we do from that we should consider all possible theories um we should consider for example when we're trying to explain something we want to know what was the call what was the origin of covet we want to know how did we get sick the other day we want to know how dangerous dangerous it is is it to move into this neighborhood um when we want to try to understand like the origin specifically like we have to consider all possible explanations because if we don't we're already setting ourselves self up for bias and self-deception so we want to be aware of the space of possibilities of of possible explanations for a certain thing when making a decision and then we want to map out the likelihood of each of those theories being true with all everything we know and we do we don't we can't do this in everyday life we don't need to we have intuition about you know what to do in a lot of simple situations but with complex situations like what person we should vote for what political candidate what political party we should um support uh i think you have to do uh this sort of bayesian analysis where you map out every possibility for example every possible theory and explanation for something you write down the evidence for each one and then you use those theories to make predictions and then you test them and then you um based on the results of those testing uh that testing update your likelihood of each of those theories being true and we're not doing this all the time so let me give you some examples um well so you know flat earthers you could you know flat earthers think the earth is flat they have these pictures where it's just flat earth uh in space and you know so if you go by this method i'm not gonna i'm not gonna rule that out immediately uh to be a good bayesian you want to consider the possibility because there's always going to be surprises in nature who would have thought quantum mechanics who would have thought superposition and entanglement was really so weird that everyone would have said that was pseudoscience they would have been like that defies the laws of physics we don't know so we consider the theory then we write down evidence for it and we try to test it so we can take a flat earth or we say we'll do a test so let's put you in a rocket we're going to pay for this we're going to fund this we're going to give you a video camera you and your fighter friends we're going to bring you up in a rocket we're going to show you we're going to look we're going to see if it's round or not everybody's trying to really figure it out because most of us haven't been to space let's say we take them into space we bring them back down they see the earth is round they come back down and they still get on their flat message boards earth flat they drugged me or something like that some impossible thing um that person is not being base optimal because they're not updating their uh model of the world um uh to be more accurate so they're going to make mistakes let me give you an example a final example of um how i can give you some new knowledge from this from a theory that will immediately change statistical behavior i want to show you the effects of like agency so let's say you're completely ignorant to whether squids or octopus are intelligent you just don't know when you're you you eat calamari at a restaurant every week sometimes you eat popcorn shrimp and then you eat calamari let's say it's randomly and you're a person that cares about you're a compassionate person so i have lunch with you and i say you know neuroscience has shown that it seems like you know squids are highly intelligent and they show all this complex behavior they have these nervous systems that would uh seem to make them intelligent um i mean lead to this like computational power where they can solve all these problems and i tell that person if they're convinced of my argument their statistical behavior is likely to change such that if you were someone and you were looking at their eating pattern all of a sudden you see them get this new knowledge they update their model and suddenly they don't eat uh the calamari anymore suddenly their appetizer becomes shrimp because they go oh that's unethical um and then maybe they're kind of weak maybe they go i love calamari and they eat it like once a month instead of like every week or something so what i'm saying is that there's a real power of knowledge that when you show the person that and they update their model there is a statistical change in behavior that's quantifiable immediately so that's the power of of of knowledge like we can we do have like agency and we can actually see that people are not behaving in a bayes optimal way and um we can see all sorts of you know disagreements about for example theories if we did a bayesian analysis we could start to converge on what is the true answer for this um so it could be a solution to the misinformation problem a solution to the misinformation problem is a solution to a problem in your own life um basically it tells you you know how you need to um update your model to be more accurate and you behave in a different way you don't have these like delusions um so you know i yeah there's uh you set goals for yourself uh in the teleological stance because basically life to stay out of equilibrium has to have these intrinsic uh survival goals like finding energy and avoiding threats but when you create a goal you you basically create an attractor and when you start working towards that goal um that is when you have a goal in your mind you say i'm going to do this for the world i'm going to build this thing that creates an attractor that you work towards if you don't create that goal in your mind you're actually not going to do what you need to do to make sure uh to work towards that happening so goal setting is absolutely something essential and if we don't do that then we're not living optimally either and actually like the law of attraction those kind of things that sound you know like woo and you're like you know but they're these best sellers why do they sell because we can actually start to map this on to the this sort of like um systems thinking and you can see oh that's why this new age stuff worked um variation selection is the method of evolution or learning or science so we can use that in our real life like if you're writing a book or something um don't just be isolated for a year writing it yourself write a little bit at a time pass it to your friends get some criticism and so that's a natural selection filter they tell you what's not good about it you go back you change it you vary it and you give it back to your reading group that's using variation and selection to optimize your writing you can use this for any project for not doing that we're making life harder um creating a collective mind so i'm saying that there's this process uh at all these hierarchical levels so when we um start to collaborate with someone we're forming a collective intelligence um that then exchanges information for us so we want to try to form collective minds and that's when you're passing your book to your friend you're doing that you're forming a collective mind and then you're doing an optimization process at the level of the collective you want to align interests so you find synergy so basically you're not you don't want to be opposed to others goals so if you all have goals you want to look at what are your similar goals and not where you're conflicting and then you will try to align those interests so we could try to do some sort of analysis with china for example where we try to uh align interest and see where we agree before you know rather than focusing on all of our disagreements and try to make progress that way and then the law of requisite variety is the final thing i'll mention right now um is that um to when you want to model the world depending on how complex your niche is you basically your model matches the complexity of the niche and so you will have a lot you will need a lot of accessible cognitive states if you're a theoretical physicist living in manhattan in this chaotic world we're living on a farm or something so you want to try to um find the niche that max matches your skill set um so you want to find low complexity niche if you're not a highly cognitive person and if you are a person that thinks a lot and your mind is just always thinking you might not want to live in a small town simple like rural town you might because you're not matched with the niche you could be highly intelligent but still you would be seen as crazy because basically um the niche is so simple that your model's too complex so um yeah there's a lot of things that come out of this that i would like to expand on but we don't have any time and we haven't really talked about free will and and and everything that we talked about at the beginning so maybe we'll have to do this again but um hopefully that was introduction to this theory of everything which creates this cosmic context that you're an agent in the theological stance is saying i'm an agent that has to survive in a world that tends towards this other disorder otherwise and to do that i have to extract energy and to extract energy i have to model the world in more detail and more sophistication as we run out of energy we're going to have to model more and more of the world so bayesian inference is how agents basically um uh carry out this process of cosmic uh complexification so life is the mechanism for complexity increase and we are agents with causal power and we have to basically um take this bayesian approach to get to the future uh that we want to see and if we don't take this approach um basically uh the world is on a sort of trajectory towards resource depletion so this is actually like becoming aware of this and this method is something that society needs to get us to the next phase transition if we don't have it we're not going to get there so what i'm saying is that this self-help system in a way there's a selection pressure that's creating it it might not be this version of it but some form of this is sort of destined to be the next big thing so everybody thinking along these lines there are other people besides me um keep going in that direction because you're onto something so yeah you're going to be the next step and then you're going to be the next step and then you're going to jump in there yet or what i think yeah bernie must have had to go wow well bobby i'll ask just two short questions from the chat just in our final minutes so first quick question from ross b what is the line between living and non-living matter how about i answer them in like a few sentences and we do a little bit more um but we'll still cut it early so the line between living and non-living matter is agency and agency comes from the causal power of information and it gets built up through an evolutionary process so you have an auto catalytic set that's a self-maintaining chemical set and there's some sort of phase transition between that chemical set and the first cell where there's a causal a change in causal structure and the agent basically gains informational control and it gains the ability to initiate new causal chains and we call that top-down causation so check out sarah walker and paul davy's work on informational control next all right very well second question from ross are these phase transitions representative of weak emergence or strong emergence in bobby's theory each new laws come into play that exert downward causation yeah i wanted to address that and that gets in the free will stuff and i would love to just come back and do a talk on free will so um what i'm saying is strong emergence has been defined wrong it's emergence that seems to violate the laws of physics the way it's been defined and what i'm saying is strong emergence is real but of course it doesn't violate the laws of physics anything that's real is part of physics so um weak emergence would be more like the formation of like a whirlpool in the bathtub like you have all these uh you have the formation of uh a coherent structure that does introduce global constraints that constrain the movement of the particles such that they're no longer chaotic they're in this ordered state but with strong emergence you actually see this change in causal structure that i just mentioned or information basically gains causal power and you have an agent

that's a control center that's actually deciding to do things and that's actually real causation
that's called top-down causation and so what strong emergence is is for example uh prior to life coming into existence you have all inanimate matter and when you have a living thing we already said you drop a rock and a pigeon off a tower one's predictable from newtonian equations one's not one you have to understand its genes and you have to understand like its relationship to its environment and then you can say oh it's flying because it has to survive and you can actually predict some things about it won't splat on the ground it's going to fly away but you can't predict it with the same thing so what i'm saying is what strong emergence is is a change in causality such that we have living agents with informational control and it's not something that's against the laws of physics or somehow working against that what it is is the laws of physics weren't complete now we're understanding that there's information that can have causal power and you do see this different type of causation
such that the living world behaves distinctly different than the non-living world
Michael the penultimate word oh that's uh put me on the spot young lads uh and again daniel you know uh what's going on in the background i'm sure people will guess but uh no um i think you know what uh bobby and
i had spoken about earlier was uh this comes down to what is our uh personal agency in terms of you know again bobby has spent a long time here and i'm gonna have to like re-listen to it myself again and uh spend some more time with it but um we're elucidating or bobby is elucidating uh what kind of agency we have
in the face of complexity um and obviously there's a what we say before bobby a methodology
there's a methodology to this uh we had some kind of catchphrase whatever but yeah so well the thing is we do have agency we behave different than inanimate systems and we do have causal power um but that's not enough that's like the first step it's like okay we're these agents then what do we do
and the the bayesian approach is kind of an instruction manual for how do you do how do you exercise this agency agency optimally and what we want to do is we want to be um aware of the uncertainty in our model and we want to try to decrease that uncertainty and we do that by for example not just having one theory about something we consider all possible theories we consider the outcomes of those theories and we make sure that we update our model uh in light of new information and if we don't do that we're not behaving optimally and we're not fully exercising our agency and you're going to see consequences of that at the societal level and so society is basically creating
pressures that are going to force us to be sort of bayesian agents because if we're not updating our model and not seeking out new information and being curious uh we're not going to be competitive
but yeah it's all about agency it's all about i will i will add to this too bob you just know to
bridge the gap between the two of you um uh daniel and pin off the word for sure on my part but uh you said act infer and serve right daniel said that i like that yeah
yeah so all this is really saying new is that active inference is not something specific to humans that organisms when you have a population of organisms and evolution is making all these copies the population is doing a sort of inference process you have these different organisms that

are different embodied models and then they're competing and then only the most accurate ones survive
so that's what bayesian inference is um and uh this should also apply at the level of society so when
you have a a social organism a society it has a model and that model is its world view it could be a
religion it could be the political model it could be authoritarianism or whatever and so with that too we
want to optimize those models and if we don't try all of the different models and and test those models
um we will have no way to know if we have the optimal model so we can't just go capitalism seems the best
or capitalism sucks let's throw it out let's do let's do communism we want to test things constantly
and update and that's error correction if we don't do that we're screwed so the bayesian the free energy
principle needs to be extended to all these levels um and when we do that then we have a whole new way to
go about optimizing social systems and political systems right now it's just like oh i think this
you know i believe in uh you know a liberal world or i believe in this conservative view and it's like
okay fine well how do we get to the bottom of it no one's going to agree first i mean let's do do
testing constant testing the proof is in the pudding it's a good idea we're going to see benefits we're
going to take surveys of everyone and get feedback and see what we can improve and um vitalio venturing
the guy who has the idea of the universe being this neural network which has been covered by media
he's explained that this multi-level learning is consistent with this neural network theory and that
you do have true emergence and you have causal power at these higher levels um and he's trying to
design a political party around this like a new approach to politics and that sort of position
that i came up with that you need to test it without me even saying anything to him that was already his
idea that he's has fleshed out you know i'm sure more than i have um so people are working on this stuff
but this is like a new niche being created right here this is the time everybody needs to go like how
can we use these systems bayesian inference and evolutionary science evolutionary theory mechanisms
to optimize society if we don't do that from if it's not entering the political discussion
we're lost we're primitive like like in a hundred years all of these things that we're talking about
i believe will be part of the discussion uh for anyone who's having any sort of ideas about how to design
systems all right bobby well you're always welcome back for 15.4 we can push into decimal places unseen
all right let's do free will let's let's bring adam and everybody back and and and do free will
free will and does voting matter two sides of the same coin yeah it is and no one's really doing that
no one's thinking that that debate has they think it's purely intellectual it's not yeah so yeah we
need to connect it to practical stuff yeah well thank you to all of our panelists and bobby as always so
both of you yeah no strengthen the journey journey well to everyone listening
all right see you later i wish yeah i wish you could hear more from brendan but uh yeah we'll follow up um
uh go to the road to omega sub stack and i'm going to post about this video and then we can have some
discussion in the comments if you see this yeah all right thank you all right thanks a lot everybody
not very well bye
thank you

Thank you.